

PANTELLERIA, Italy

WMO: 164700

Lat: 36.814N

Lon: 11.971E

Elev: 650

StdP: 14.35

Time Zone: 1.00 (EUC)

Period: 86-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	44.4	46.3	27.9	22.4	60.3	31.8	26.9	56.0	43.8	52.2	38.1	51.9	14.4	250	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	93.5	69.7	89.6	68.7	86.3	69.9	79.8	83.6	78.0	81.9	76.6	80.6	13.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	153.4	82.4	76.9	143.6	80.9	75.3	135.8	79.4	43.9	83.5	42.0	81.7	40.4	80.9	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.4	30.0	26.1	DB	41.7	99.4	3.6	4.6	39.1	102.7	37.0	105.4	35.0	108.0	32.4	111.4	(4)
(5)			WB	37.3	83.3	2.8	2.6	35.3	85.2	33.6	86.7	32.1	88.1	30.1	90.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.3	53.9	53.6	55.9	59.5	65.8	72.8	77.7	79.0	75.1	70.5	62.3	56.0		
	DBStd	10.19	3.32	3.63	4.59	4.12	6.03	6.01	4.77	3.54	3.93	4.08	4.28	4.14		
	HDD50	29	7	10	7	0	0	0	0	0	0	0	0	5		
	HDD65	1569	343	319	285	174	57	3	0	0	0	4	103	281		
	CDD50	5599	129	110	190	285	491	684	859	900	754	637	369	191		
	CDD65	1666	0	0	3	9	83	237	394	435	305	176	22	2		
	CDH74	11052	0	0	5	37	512	1593	3127	3662	1629	473	14	0		
	CDH80	3128	0	0	0	5	187	565	1003	1004	318	46	0	0		
(14) Wind	WSAvg	11.6	12.4	13.0	13.3	13.4	11.9	9.7	9.7	10.1	10.4	10.9	11.9	13.0		
(15) Precipitation	PrecAvg	23.3	3.5	2.4	2.0	1.5	0.6	0.3	0.1	0.2	2.9	3.1	3.3	3.6		
	PrecMax	50.6	11.2	6.8	5.9	5.4	2.2	3.9	1.5	1.6	17.9	7.8	11.0	10.2		
	PrecMin	10.9	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.4	0.3		
	PrecStd	7.4	2.9	1.7	1.6	1.1	0.6	0.8	0.3	0.4	3.6	1.7	2.0	2.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.0	67.6	73.8	78.7	92.7	99.4	99.9	96.7	91.8	85.3	76.0	69.6		
		MCWB	52.9	53.0	55.1	60.9	70.9	73.1	73.0	70.8	67.3	67.9	62.1	60.1		
	2%	DB	61.8	62.8	68.1	73.0	86.1	91.3	93.5	91.3	86.2	80.7	72.6	66.5		
		MCWB	54.6	53.9	53.5	58.4	64.2	68.5	68.8	70.0	69.3	68.0	64.5	60.3		
	5%	DB	59.3	60.1	64.6	69.2	79.2	85.8	89.0	87.8	82.9	78.5	70.0	64.0		
		MCWB	54.6	54.7	53.7	57.5	62.8	65.7	69.8	71.3	71.8	69.0	63.5	59.0		
	10%	DB	57.9	58.1	61.7	66.1	75.1	82.3	85.6	85.2	80.7	76.7	68.1	61.1		
		MCWB	54.0	54.1	54.6	57.3	61.8	66.0	71.2	72.4	71.8	69.0	62.8	57.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.3	58.6	60.8	64.7	73.6	76.9	82.7	82.7	79.0	76.3	69.9	63.6		
		MCDB	60.7	61.4	64.6	71.4	84.5	82.5	85.6	86.4	82.0	79.3	72.5	67.3		
	2%	WB	56.8	56.8	58.4	62.4	69.7	74.2	79.9	80.1	77.2	73.5	66.5	61.1		
		MCDB	59.6	60.2	62.4	67.6	76.2	80.3	83.6	83.8	80.8	76.6	69.5	64.7		
	5%	WB	55.6	55.6	57.2	60.6	67.1	72.5	77.9	78.2	75.5	72.0	65.1	59.4		
		MCDB	58.4	58.7	60.7	65.2	74.3	78.7	81.9	81.9	79.3	75.8	68.4	62.5		
	10%	WB	54.5	54.5	56.0	59.1	64.9	70.9	76.1	76.9	73.9	70.7	63.6	58.0		
		MCDB	57.4	57.4	59.3	63.6	71.1	76.6	80.4	80.7	78.0	74.9	67.1	60.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.5	7.5	9.0	10.1	11.0	9.8	9.4	8.8	8.1	6.4	5.7		
		MCDBR	7.9	9.0	12.4	14.9	16.7	17.3	15.5	14.5	12.3	10.9	8.1	7.9		
	5% WB	MCWBR	7.1	6.8	9.4	11.2	10.7	11.2	12.9	12.8	11.5	9.5	7.6	7.4		
		MCWBR	6.8	7.3	8.6	11.6	13.7	12.8	10.5	10.4	9.4	7.2	6.2	6.2		
(40) Clear-Sky Solar Irradiance	taub		0.343	0.362	0.408	0.448	0.455	0.426	0.440	0.437	0.439	0.413	0.377	0.351		
	taud		2.483	2.414	2.266	2.168	2.190	2.319	2.287	2.308	2.301	2.341	2.404	2.456		
	Ebn at Noon		267	275	271	266	265	272	267	266	258	253	251	256		
	Edh at Noon		26	32	40	47	46	41	42	40	38	34	28	26		
(44) All-Sky Solar Radiation	RadAvg		763	1043	1465	1866	2195	2429	2452	2173	1633	1219	860	686		
	RadStd		56	83	102	102	114	91	56	75	73	70	42	55		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page